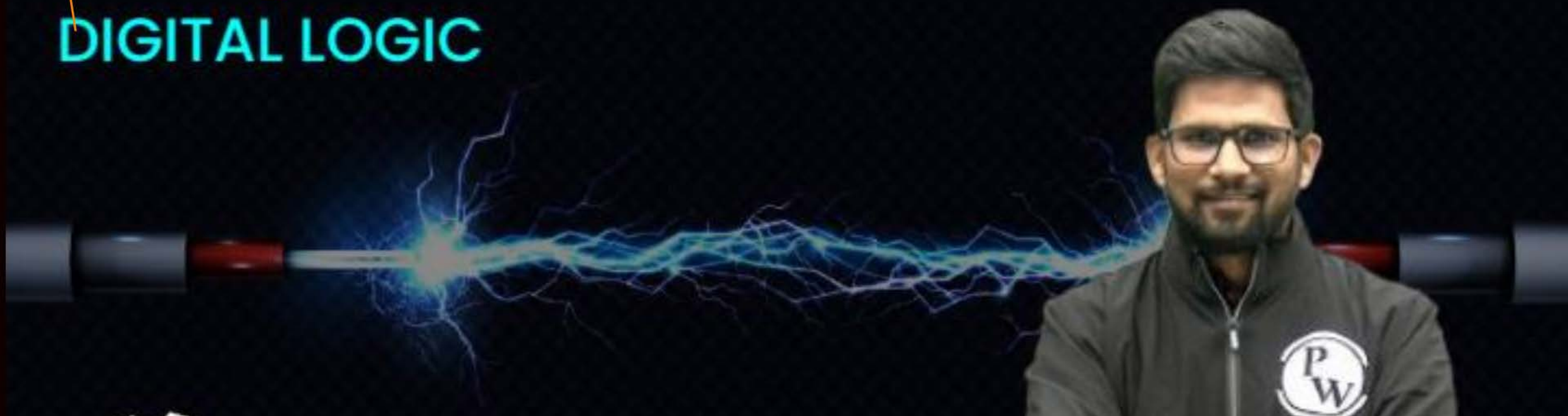
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 04

Miscellaneous Topics



By- Chandan Gupta Sir

Recap of Previous Lecture



- Number System
-
-



Number System Cont.

Question

Two numbers A and B are represented using 2's compliment representation :

$$A = (1011)_2, \qquad B = (1110)_2$$

Then the results of (B – A) will be

(a) $(0011)_2$

(b) $(10011)_2$

(c) $(1000)_2$

(d) $(1101)_2$

$A = (1011)_2 = (-5)_{10}$

$B = (1110)_2 = (-2)$

$B - A = (3)_{10}$

$B = 1110$

$-A = 0101$

$+$

$\leftarrow \underline{\textcircled{1}0011} \longrightarrow$ Result $= (0011)_2 = (+3)$

discarded

Question

Three numbers A, B and C are represented using 2's compliment representation as :

$$A = (1010)_2, \quad B = (0111)_2, \quad C = (1100)_2$$

Then the results of (A + B – C) in 2's compliment representation as :

(a) $(0011)_2$

(b) $(0101)_2$

(c) $(1011)_2$

(d) $(1101)_2$

$A+B \Rightarrow$

$$\begin{array}{r} 1010 \\ +\ 0111 \\ \hline (1)\ 0001 \end{array}$$

(1) discarded

Result $0001 = (+1)_{10}$

$(A+B)-C \Rightarrow$

$$\begin{array}{r} = 0001 \\ +\ 0100 \\ \hline 0101 \end{array}$$

Result $= (0101) = (+5)_{10}$

Question

A signed no. is represented using 2's compliment representation as :

$N_1 = (1001)_2 = (1111\ 1001)_2$

Then its 8-bit representation will be :

(a) $(1001\ 1001)_2$ ✗

(b) $(1111\ 1001)_2$ ✓

(c) $(1000\ 1001)_2$ ✗

(d) $(0000\ 1001)_2$ ✗

Question

A 4-bit number in 2's compliment sign no. representation is

M = | S_3 | S_2 | S_1 | S_0 | $= (0001)_2 = (1)_{10}$

and other no. given in 2's compliment representation is

N = | S_3 | S_3 | S_2 | S_1 | S_0 | 1 | 0 |

Then the relation between M and N is :

(a) N = 4M

(b) M = 4N + 2

(c) N = 4M + 2

(d) N = 4M + 1

$N = 0000110 = (+6)$

$N = 4M + 2$

$M = S_3 S_2 S_1 S_0$

$M = S_3 S_3 S_2 S_1 S_0$

$2M = S_3 S_3 S_2 S_1 S_0 0$

$2 \times 2M = S_3 S_3 S_2 S_1 S_0 0 0 = 4M$

$+ 0000010 = 2$

$N = S_3 S_3 S_2 S_1 S_0 1 0$

$1010 = (-6)_{10}$

$10100 = (-12)_{10}$

$101000 = (-24)_{10}$

$0100 = 4$

$01000 = 8$

$010000 = 16$

BCD Codes

→ Weighted code system

8421 codes

	8 4 2 1
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1

→ it is not a self complimentry.

→ Valid BCD Codes

1 0 1 0
1 0 1 1
1 1 0 0
1 1 0 1
1 1 1 0
1 1 1 1

→ invalid BCD Codes

$(25)_{10} = (11001)_2 = (0010\ 0101)_{BCD}$

$(3586)_{10} = (0011\ 0101\ 1000\ 0110)_{BCD}$

Excess - 3 Codes

→ Unweighted Code system

Invalid Excess-3 codes
0000, 0001, 0010, 1101,
1110, 1111

Note: Excess-3 code system is the only unweighted code system which is self complimentry in nature.

	BCD code	Excess-3 code
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

→ Self complimentry code

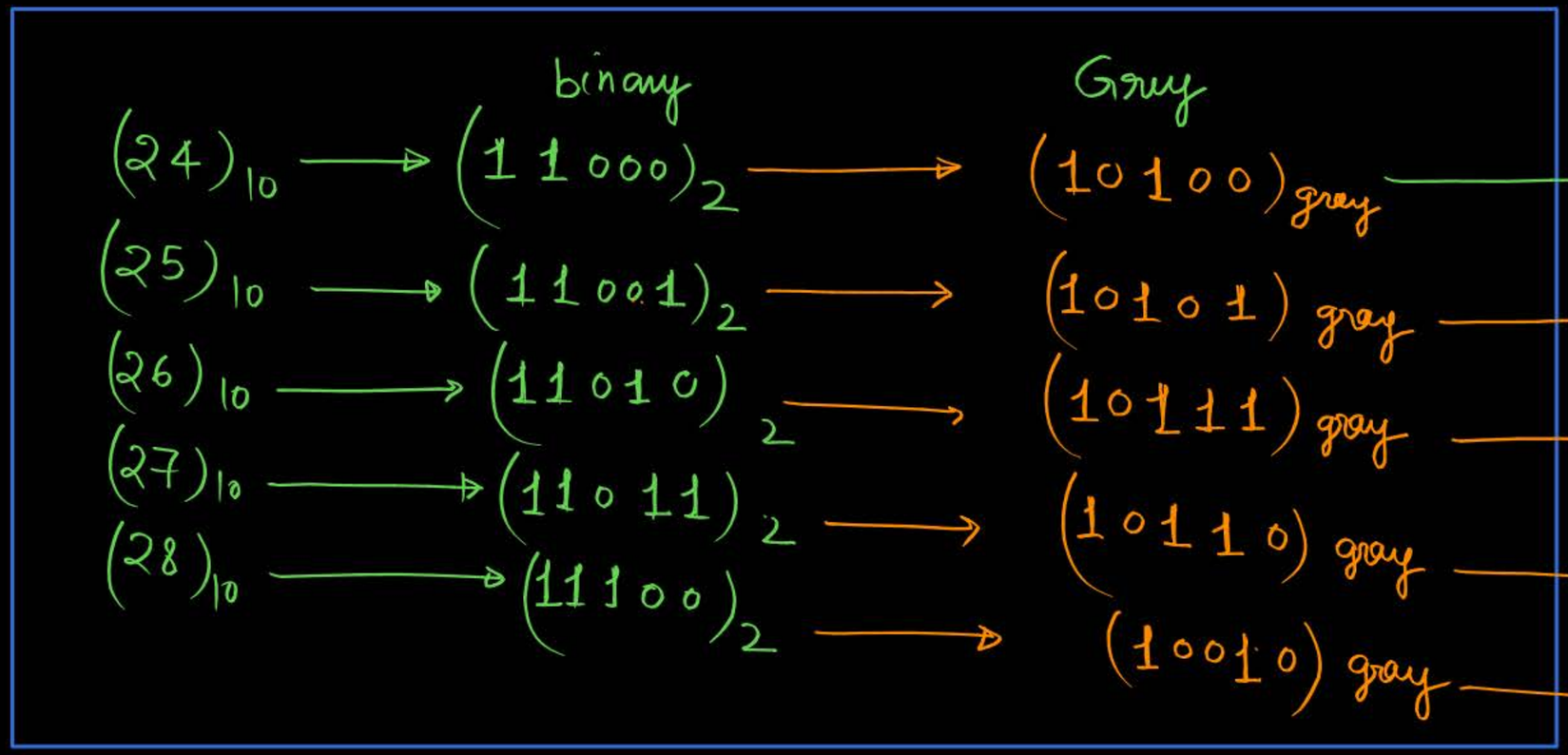
Weighted code system having summation of weights equal to 9 are self complimentry in nature.

→ 2421, 4311, 4221, etc.

- $(35879)_{10} = (0011\ 0101\ 1000\ 0111\ 1001)_{BCD} = (0110\ 1000\ 1011\ 1010\ 1100)$ Excess -3.

Gray Code / Number System

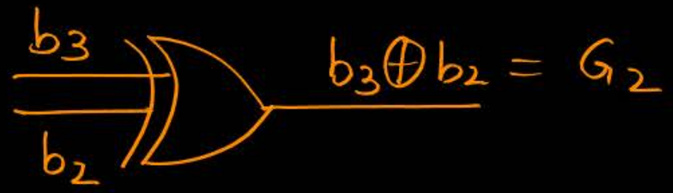
- Unweighted Number System
- Unit distance code

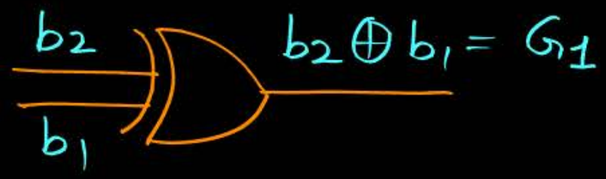


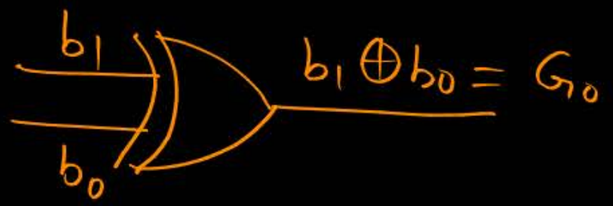
Binary to Gray Code Conversion :

$b_3\ b_2\ b_1\ b_0 \longrightarrow G_3 G_2 G_1 G_0$

$b_3 = G_3$







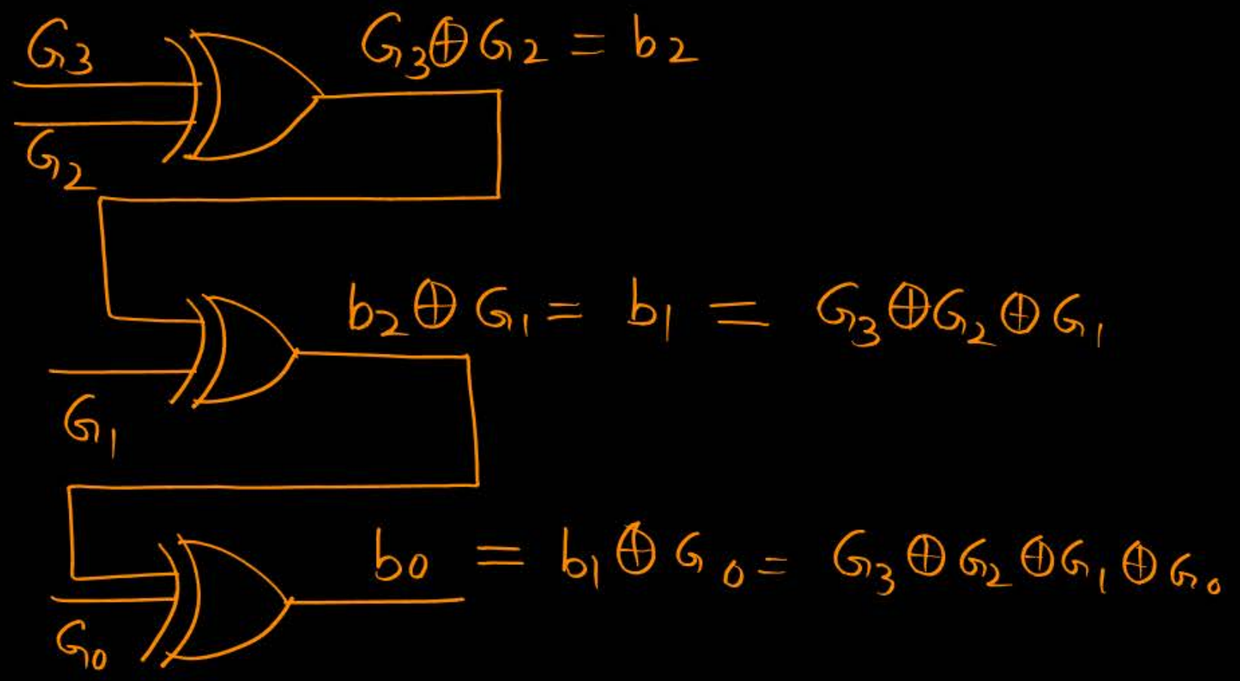
$(136)_{10} = (10001000)_2$

$= (11001100)_{gray}$

Gray to Binary Conversion :

$G_3 G_2 G_1 G_0 \longrightarrow b_3 b_2 b_1 b_0$

$G_3 = b_3$





Topic : 2 Min Summary

Number System

Thank you

GW

Soldiers !

